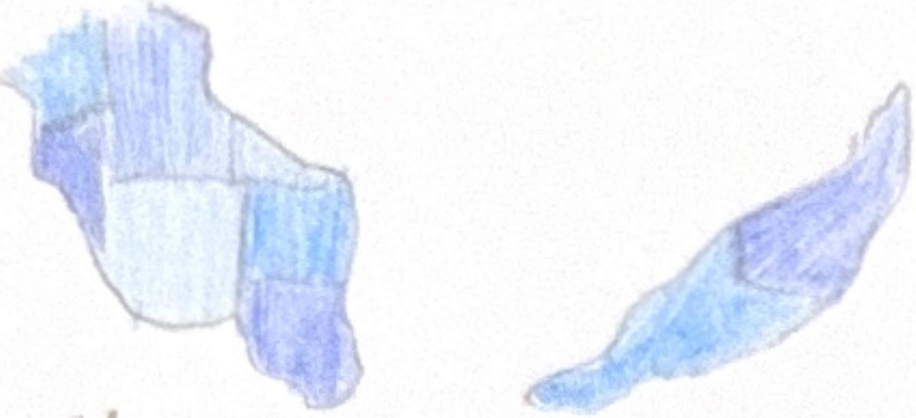
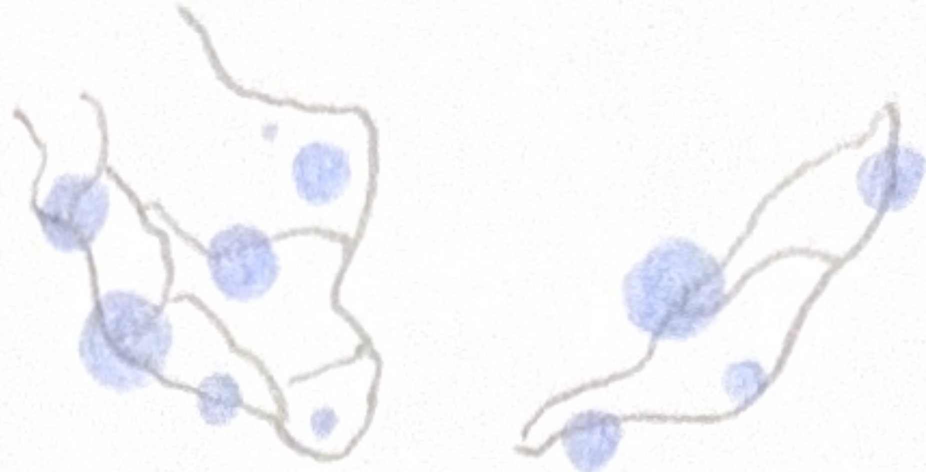
1. IDEAS

a. temperature / pollution level / wind speed with map

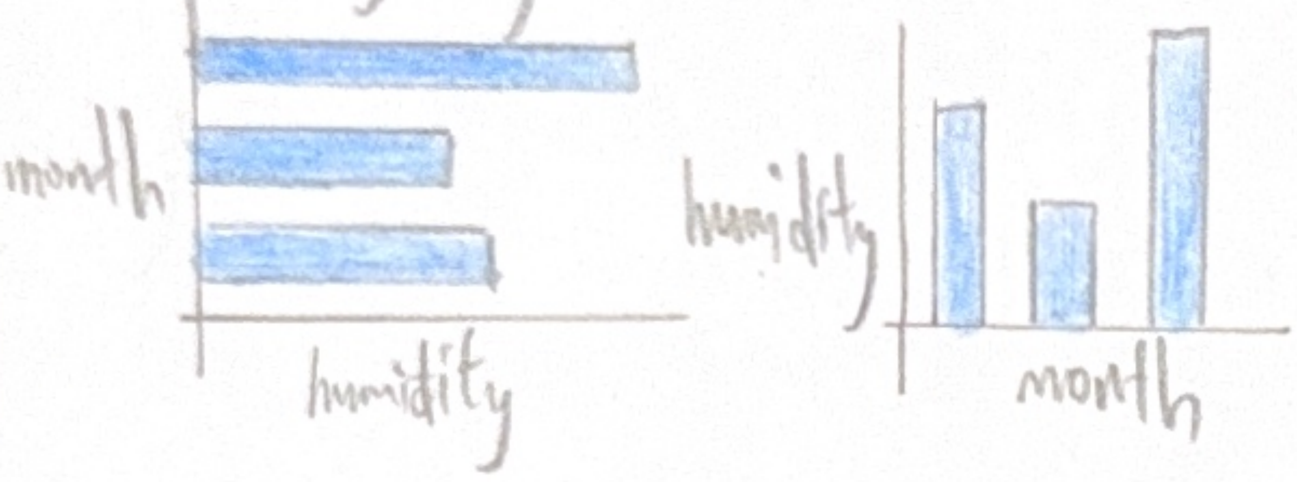


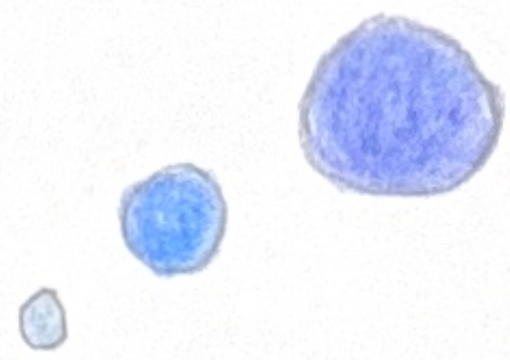
choropleth map



proportional symbol map

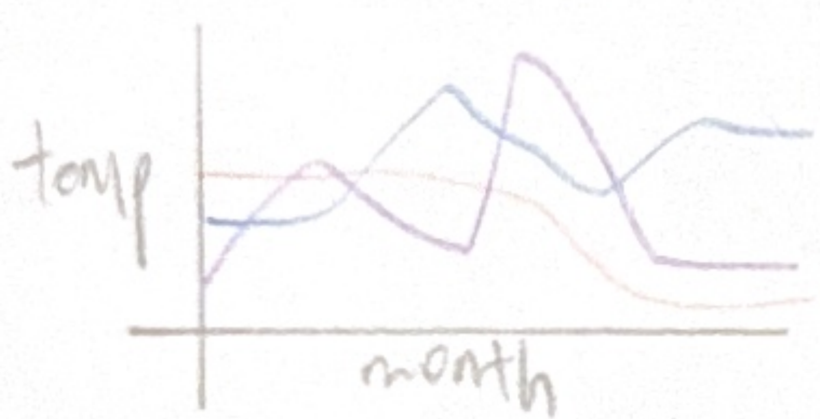
b. humidity by month



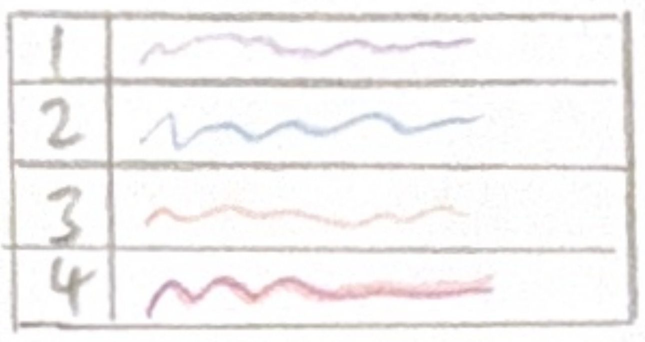


ordered proportional symbol

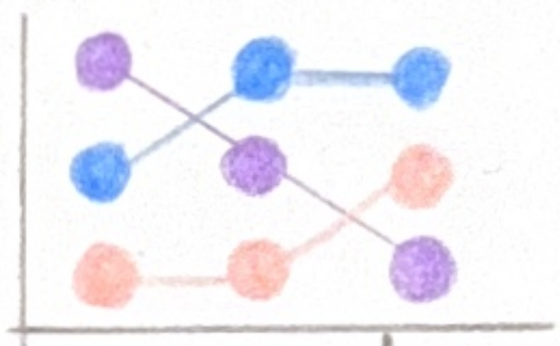
c. temperature by month



multi-line chart

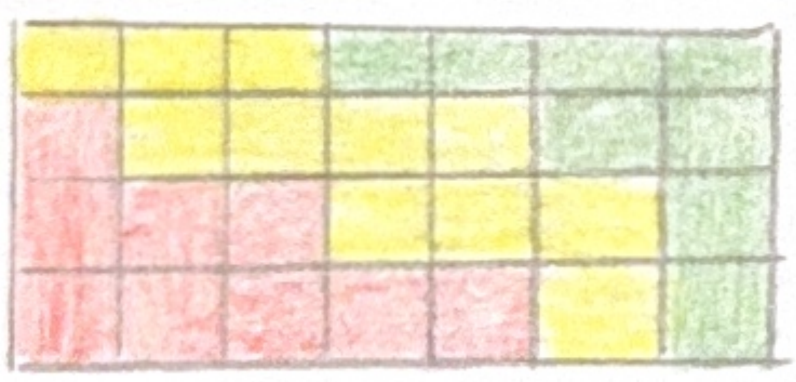


rankings



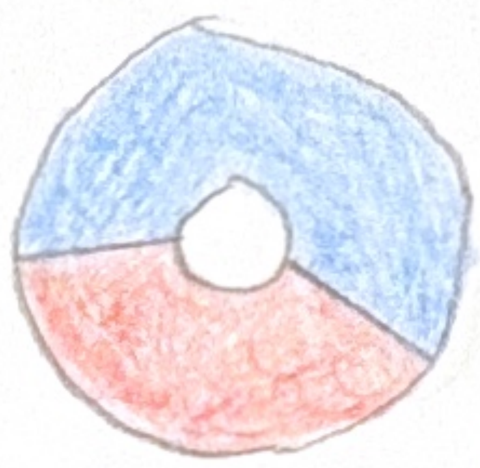
bump chart

d. month x state for attribute



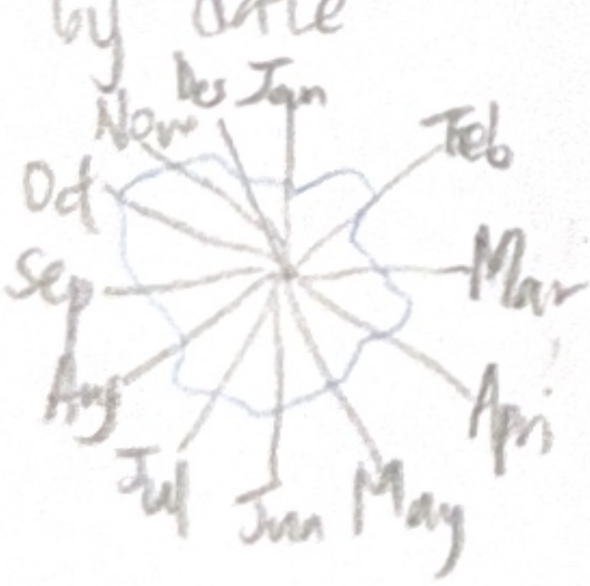
xy heat map

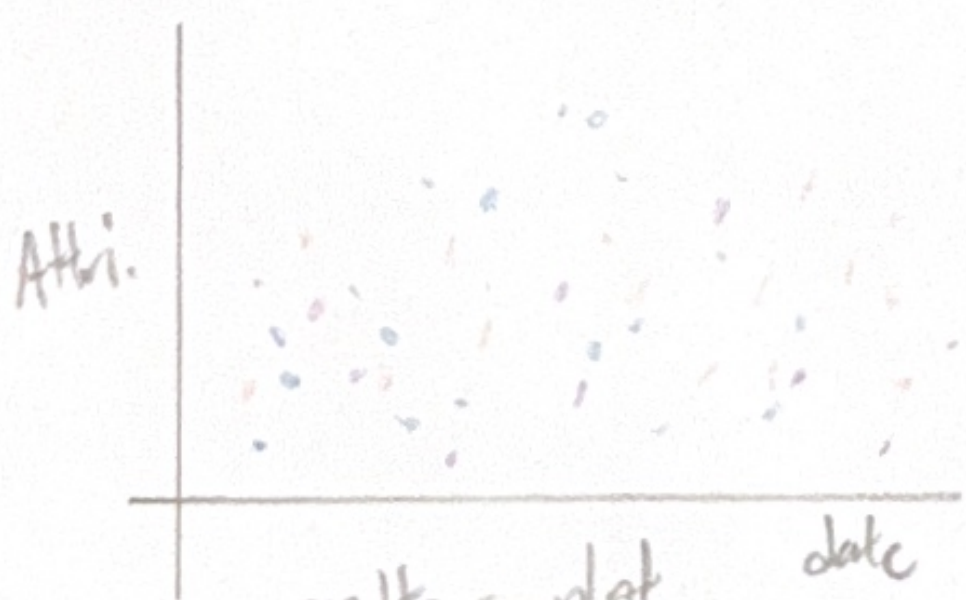
f. east vs south malaysia



donut chart

e. by date





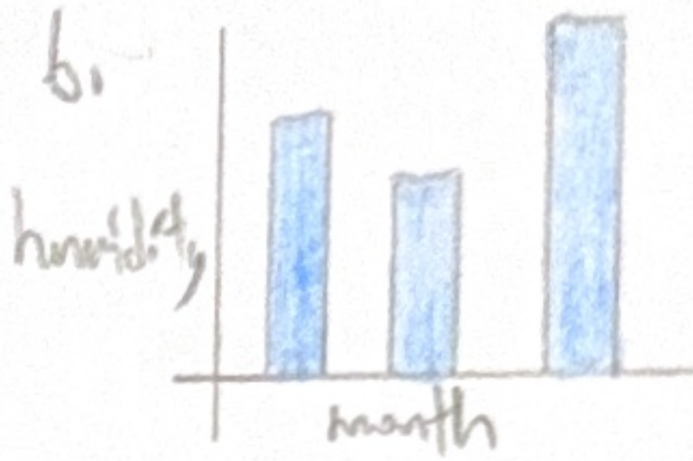
scatter plot

2. FILTER

a.

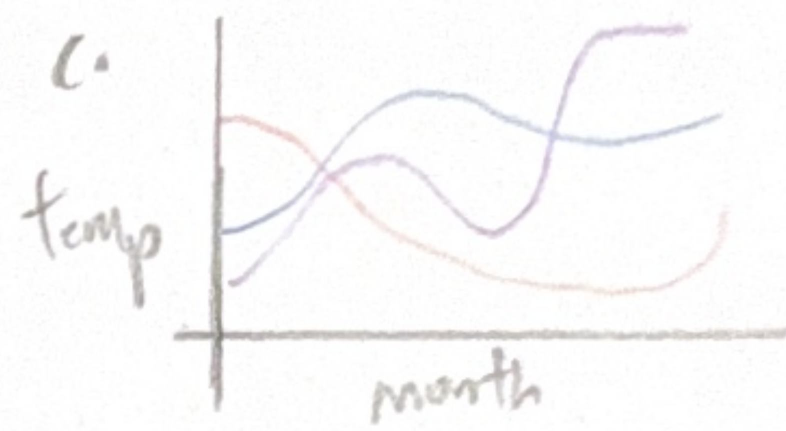
Better choice since the air pollution level is by station.

b.



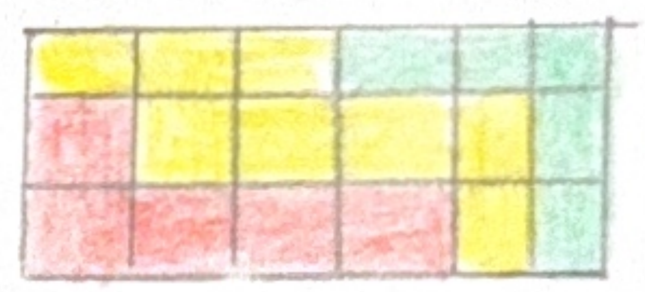
Better visualisation for quantitative data

c.



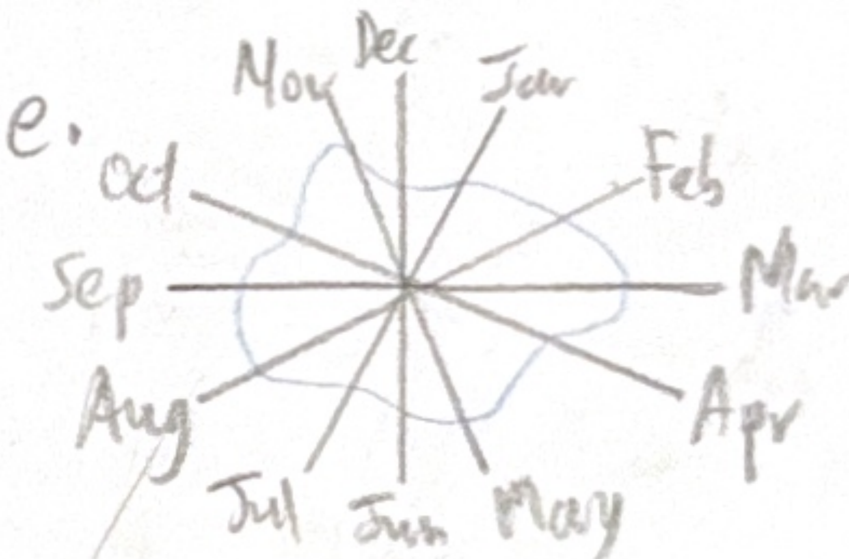
Better show the trend of temp

d.



Best to show both month and state aspect

e.



Show trend over Month across year

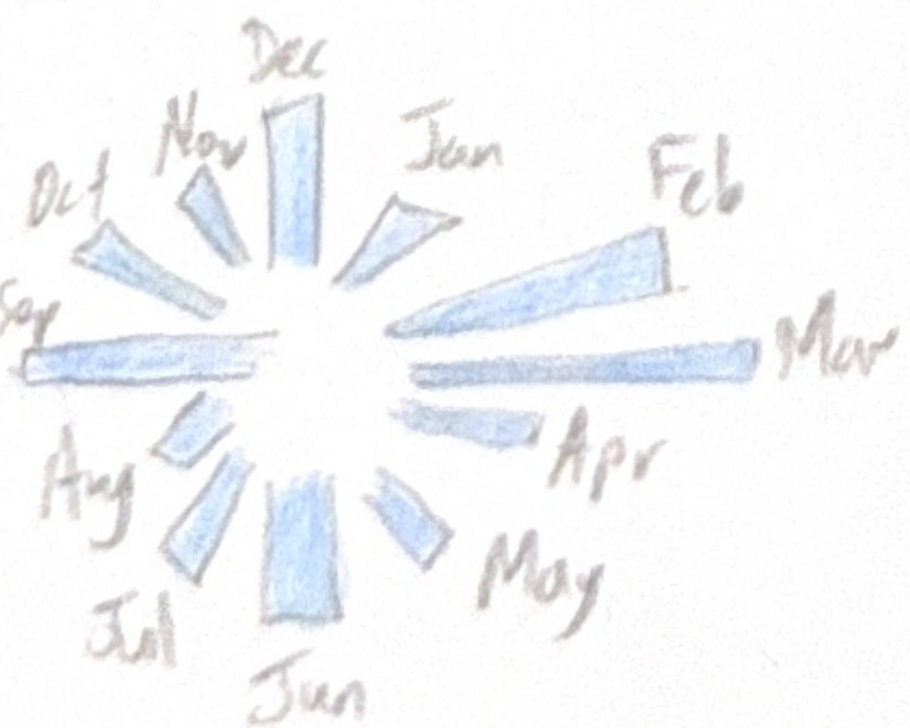
3. CATEGORIZE

Map show pollution level → Visual the temperature by state

↓

Show wind speed by heat map ← Show the trend of humidity across year

4. COMBINE AND REFINE

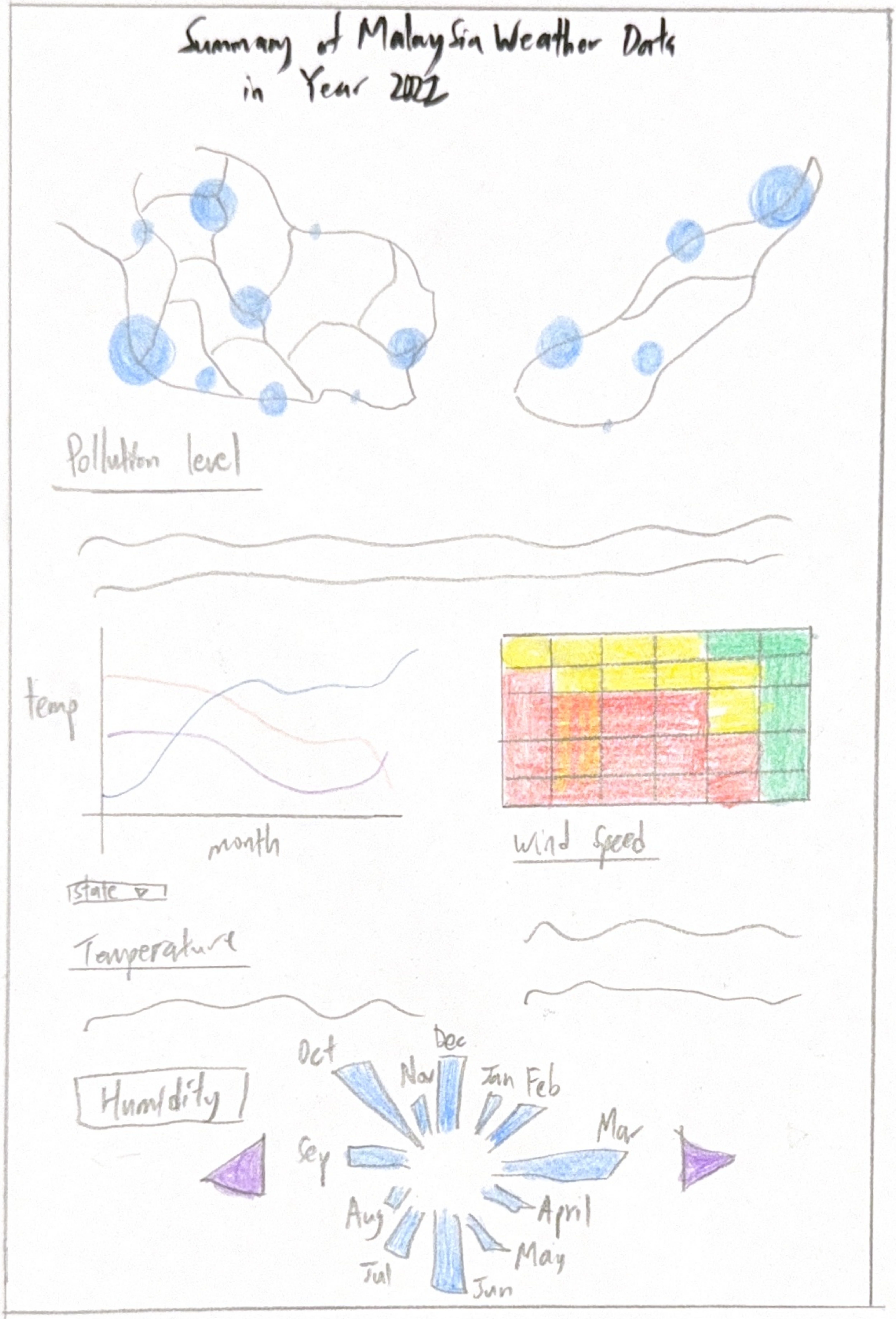


- Refine bar chart to show month trend
- Avoid messy data visualisation

5. QUESTION

1. Does the visualisation show the pollution level of each district?
2. Does this visualisation show the trend of temperature in each state?
3. Does this visualisation show the trend of humidity in each month?
4. Does this visualisation show wind speed of each state?

LAYOUT



Title : Summary of Malaysia weather Data Year 2022
Author : Nigel Tan Yu Zhe
Date : 17/10/2025
Sheet : 2
Task : Initial design of visualisation

OPERATION

There are Dropdown Menu to choose the state for temperature so that it can filter out specific line.
Help user to see the line they want to focus on.

Tooltip will be used to show more detail information for each graph, so that they know the actual value.

DISCUSSION

Adv.

1. Well divided to different section
2. Allows interactivity to choose state user want to focus
3. Can change graph without much moving of graph.

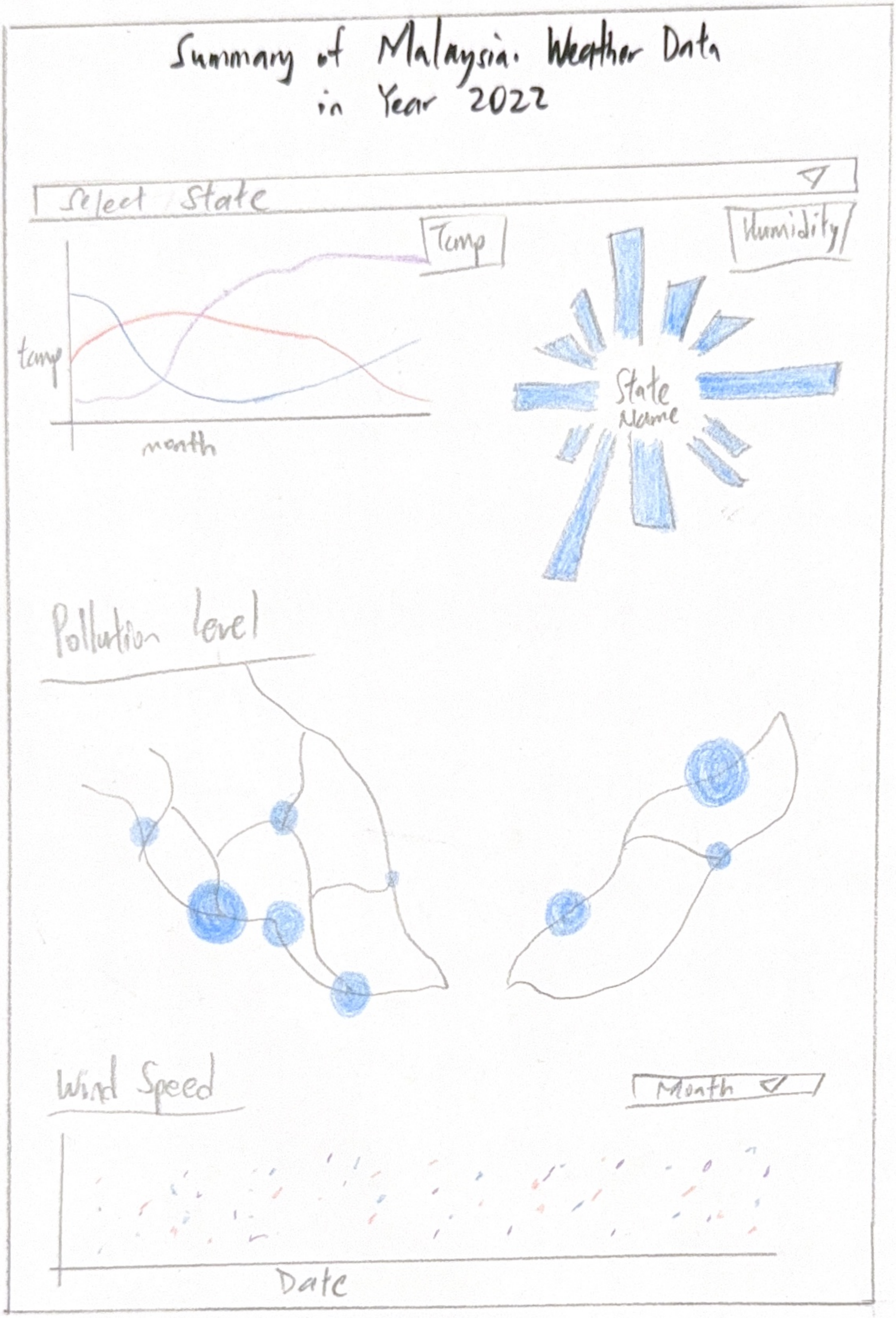
Disadv.

1. Require short term memory
2. Hard to get main insight

FOCUS

- No Main Focus, all graph are important. But some graph will be larger to emphasize the hierarchy
- High data ink-ratio as all data is presented by graph, less text used
- Equally divided into 3 section, so clearer view of presentation.
- Can specify the color used for each state, so that every graph color used is representing same state.

LAYOUT



Title : Summary of Malaysia Weather Data Year 2022
Author : Nigel Tan Yu Zhe
Date : 17/10/2025
Sheet : 3
Task : Initial design of visualisation

OPERATION

There are state dropdown menu to filter state for temp and humidity.
Help user to see the trend for specific state.
Month dropdown menu to see the highest wind speed in each day.
Tooltip will be used to show more detailed information for each graph.

DISCUSSION

Adv.

1. Clean layout, separate to different section
2. Support lots of user interactive elements
3. Can change graph without much change

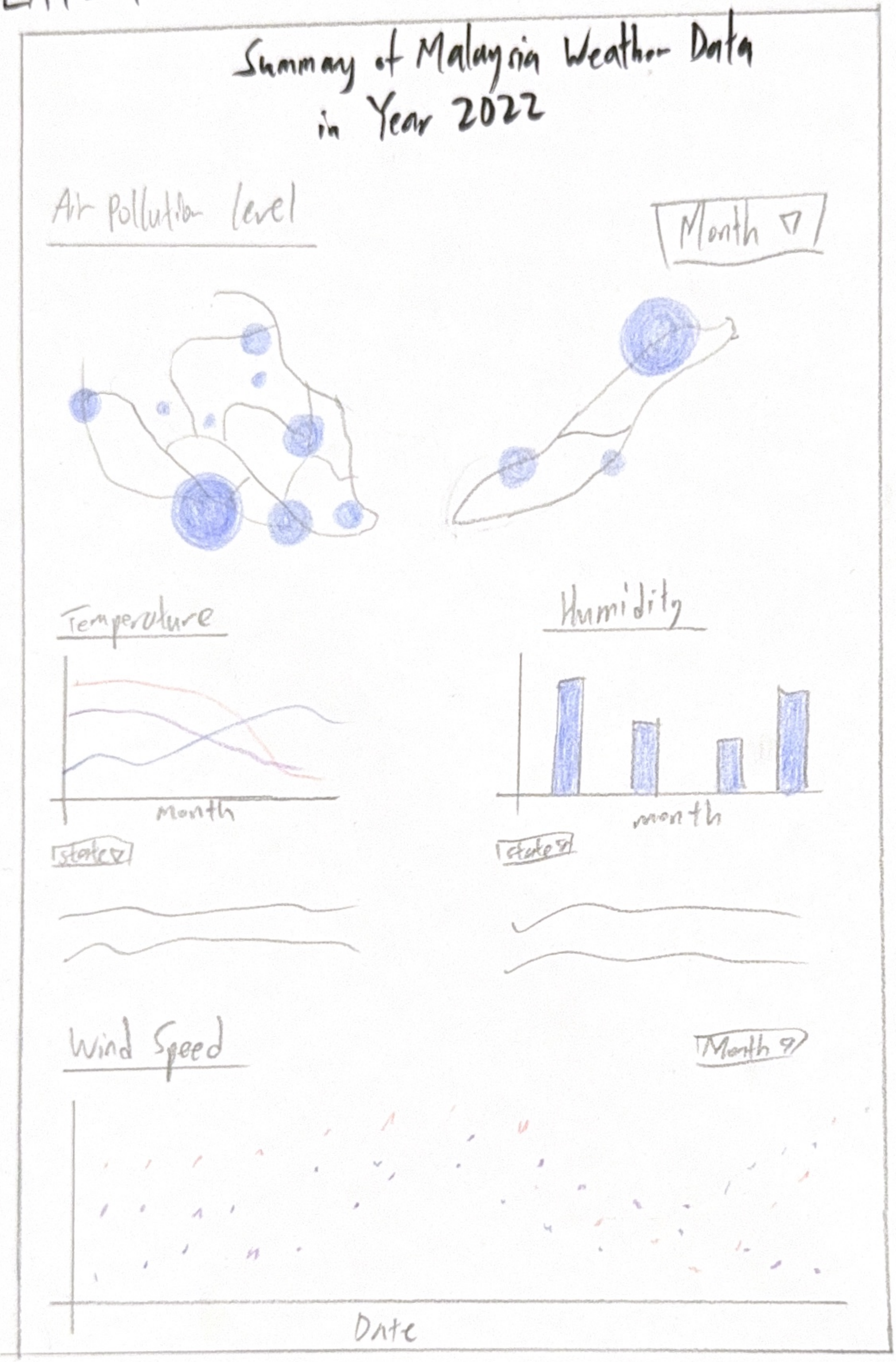
Disadv.

1. Require short memory
2. Less insight no paragraph explanation

FOCUS

- Will focus the proportional symbol map as the map placed in visual centre
- A filter control both graph at top of visualisation, giving comprehensive information
- Divided into 3 section, each with different aspect to the attribute finding
- Specify colour for each state, so that colour represent them well and continuously
- No text paragraph, use graph that is self explanatory.
- More user interaction compare to other visualisation layout.

LAYOUT



Title : Summary of Malaysia Weather Data Year 2022

Author : Nigel Tan Yu Zhe

Date : 17/10/2025

Sheet : 4

Task : Initial design of visualisation

OPERATION

There are state dropdown menu to filter state for tempt and humidity.

Help user to see the trend for specific state.

There are month dropdown menu to filter month for wind speed and pollution.

Help user to see the trend for specific month

DISCUSSION

Adv :

1. Many user interactivity options to filter data
2. Can change graph without much change.

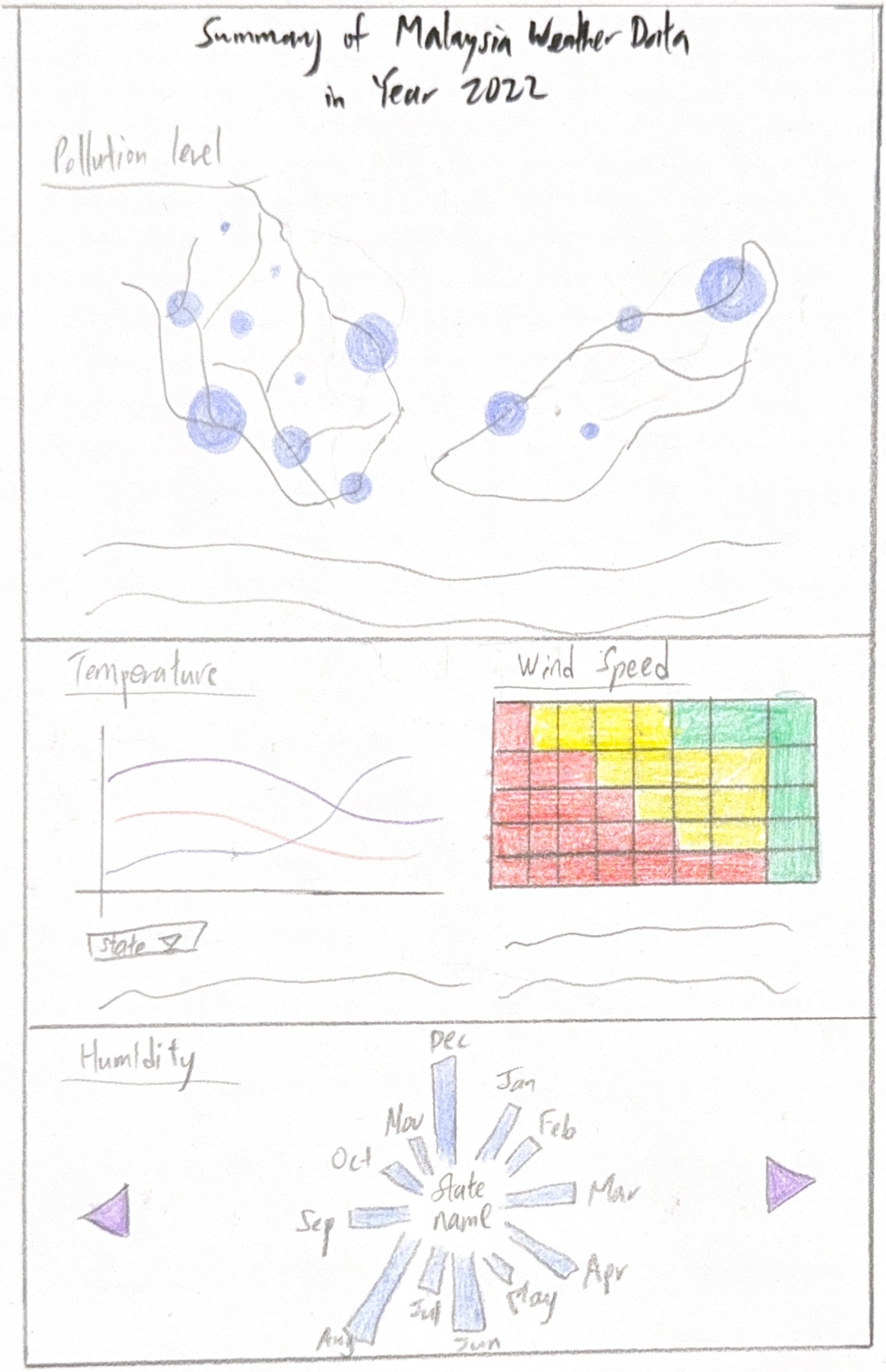
Disadv :

1. Require lots of short term memory
2. less paragraph to explain graph

FOCUS

- No map focus, all graph are important. But some graph will be larger to emphasize.
- High data ink-ratio, less paragraph is used
- Equally divided into 3 sections, so clearer view of presentation.
- Can specify color use for each state, so that every graph colour used is representing same state

LAYOUT Based on sheet 2



Title : Summary of Weather Data Year 2022
Author : Nigel Tan Yu Zhe
Date : 17/10/2025
Sheet : 5
Task : Final design of visualisation

OPERATION

There are state dropdown menu to choose the state for temperature so that it can filter out specific line.
Help user to see the line they want to focus on.
Left right button to change state to see the humidity
Tooltip is used to give extra detail information

[redacted] DETAIL

1. This will be a scroll-able visualisation top to down
2. Mostly complete by vegalite And use R for data aggregation.
3. To build this visualisation 1 week is needed
4. Need a desktop or laptop to run data - pre processing
5. Topojson is needed for map

FOCUS

- Main Focus is the proportional symbol map which have bigger size.
- High data ink ratio, less paragraph text as the graph is self explanatory
- Can specify color used for each state, so that every graph colour used is representing same state
- Have lots of interactive elements to control which aspects and to better filter them out.